

Toolchain

A **toolchain** is a set of software development tools used to build and otherwise develop software. Often, the tools are executed sequentially and form a pipeline such that the output of one tool is the input for the next. Sometimes the term is used for a set of related tools that are not necessarily executed sequentially.^{[1][2][3]}

A relatively common and simple toolchain consists of the tools to build for a particular operating system (OS) and CPU architecture: a compiler, a linker, and a debugger. With a cross-compiler, a toolchain can support cross-platform development.

For building more complex software systems, many other tools may be in the toolchain. For example, for a video game, the toolchain may include tools for preparing sound effects, music, textures, 3-dimensional models and animations, and for combining these resources into the finished product.^{[1][2]}

See also



- Buildroot – Tool for building Linux
- Debian build toolchain – Set of programming tools for deploy packages for Debian repositories
- DevOps toolchain – Tools for software development
- Framework – Software that supports solution development via inversion of control
- Library – Collection of resources used to develop a computer program
- GNU toolchain – Collection of programming tools produced by the GNU Project
- LLVM – Compiler backend for multiple programming languages

References

1. "Toolchain Overview" (<http://www.nongnu.org/avr-libc/user-manual/overview.html>). *nongnu.org*. 2012-01-03. Retrieved 2013-10-21.
2. "Toolchains" (<http://elinux.org/Toolchains>). *elinux.org*. 2013-09-08. Retrieved 2013-10-21.
3. Imran, Saed; Buchheit, Martin; Hollunder, Bernhard; Schreier, Ulf (2015-10-29). *Tool Chains in Agile ALM Environments: A Short Introduction*. Lecture Notes in Computer Science. Vol. 9416. pp. 371–380. doi:10.1007/978-3-319-26138-6_40 (https://doi.org/10.1007%2F978-3-319-26138-6_40). ISBN 978-3-319-26137-9.